

AXIOM NETWORK PROTOCOL LAYER

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Sovereign AI Training Data Layer

PROTOCOL THESIS & MECHANICS v2.4

Abstract: A decentralized peer-to-peer data alignment network replacing manual consulting overheads with math. By fractionalizing curated domain expert datasets into cryptographic ledger pools, Axiom introduces structural reward multipliers and perpetual compounding royalties back to contributing expert validator nodes.

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Deployment Target: Sovereign Mainnet v2

1. Abstract & Vision

The rapidly escalating performance demands of generative artificial intelligence model architectures (such as Large Language Models, Multi-Modal Systems, and Reinforcement Learning from Human Feedback Agents) are heavily constrained by the quality and authenticity of their alignment data. Traditional crowdsourced annotation structures have failed, resulting in systemic hallucination reinforcement and high manual engineering overheads.

Axiom replaces administrative consultancy overheads with math. By fractionalizing curated domain expert datasets into cryptographic ledger pools, Axiom introduces a peer-to-peer training network that provides perpetual compounding royalty dividends back to the expert validators who created and verified the data.

By routing multi-turn annotation requests through double-blind validator structures and calculating consensus validation using advanced margin checks, Axiom enforces extreme dataset fidelity. This produces pre-training and fine-tuning datasets that achieve higher downstream alignment benchmarks than conventional service consulting firms at a fraction of the cost.

2. Relational Royalty Model

The core value vector of Axiom is the Relational Royalty Engine. Unlike standard crowdsourcing platforms where expert annotations are bought once as flat commodities, approved expert validations inside Axiom remain perpetually linked to their contributors.

Compounding Yield Mechanics:

A. Upfront Payout Vector

Experts claim and resolve tasks inside specific domain vetting pools, instantly earning baseline upfront points. These points are mapped directly into payouts via integration with payment rails (Razorpay/Stripe).

B. Passive Royalty Stream

Whenever an enterprise purchaser licenses that specific asset pool to reinforce their LLMs, secondary marketplace royalties (typically averaged at 5% for pools) are routed directly back to the original validators pro-rata.

This creates an asset-class structure out of human-labeled inputs. The Relational Royalty formula integrates complexity factors, active vetting tier multipliers, and double-blind validation ratings to distribute royal dividends proportionally and mathematically back to contributing expert nodes.

3. Cryptographic Consensus QA Engine

To eliminate visual inspect pipelines and manual administrative audits, Axiom manages annotation accuracy using a randomized multi-model cryptographic consensus checker.

Double-Blind Verification Protocol:

Every submission submitted by a specialist contributor is immediately routed to a queue in the background. Two independent vetting nodes (validators) verify the submission blindly. If the scoring margin matches standard quality parameters (e.g. >95%), the task is instantly approved.

In cases of a discrepancy, the task is routed to a third-tier consensus audit. Our consensus algorithms calibrate margin score boundaries across LLMs (using Groq Llama 3.3 and GPT-4o validations) to automatically slash rewards for poor annotation nodes, guaranteeing clean data generation pipelines.

4. Vetting Arena & Node Tiers Math

Axiom contributors are not anonymous crowdsourcers; they are vetted domain expert nodes. To qualify for specific asset pools, contributors must enter the Vetting Arena, passing automated time-based challenges mapped directly to their domains.

Based on their performance in the Vetting Arena, nodes receive structural tiers that multiply their point gains:

Node Tier Level	Min XP	Point Multiplier	Access Priority
Bronze Triage	0 XP	1.0x	Standard Queue
Silver Triage	1,000 XP	1.2x	Priority Access
Gold Triage	5,000 XP	1.5x	High-Yield Pools
Senior Triage	15,000 XP	1.7x	Consensus Auditor
Elite Triage	50,000 XP	2.0x	System Genesis